STA5076Z – Supervised Learning 2026
Assignment 2

Due Date: Wednesday 13 May 2026, 16:00

Please read the following instructions carefully:

- Answer all questions using R; you may use any packages and functions you like.
- Present your report in a **pdf file**, which is to be submitted to **Assignment 2 – Write-up** on Amathuba.
- Your pdf report may be compiled using any software you wish, although you are strongly encouraged to use R Markdown/Quarto or LaTeX. Some marks will be awarded for communication and presentation, so make sure your report is neat.
- Upload all relevant R files (.R, .Rmd, .qmd, .Rdata, etc.) as well as your test set predictions to **Assignment 2 – Code & Predictions**. **Please do not zip these files**; upload them separately.
- Set seeds appropriately to ensure that all results are exactly reproducible.
- You are encouraged to include figures and tables, but only if they are relevant to your discussion. Use descriptive captions! One should be able to understand what a figure represents or a table contains based on the caption.
- Also be sure to interpret figures sufficiently; marks will be deducted for any output placed in the report without discussion.
- Structure and write your report according to the sections given below.
- Do NOT paste R output verbatim; this will be penalised. If you want to include R output, typeset it properly or present it in a table.
- To help the reader easily assimilate the information, round values to the fewest number of decimal places necessary (unless there is a reason for expressing a more exact value).
- The report itself should NOT contain any code – use `echo = FALSE` for .Rmd/.qmd files and do NOT attach any code as an appendix.
- Reference your sources appropriately.
- **The page limit is 10 pages**. Anything beyond the 10th page will be ignored.
- You are expected to work on this **on your own**. Note that your submission will be checked for plagiarism – both from other submissions and online resources – by Turnitin.

Cape Town Airbnb listing analysis

It is well documented that short-term rental platforms such as Airbnb have had significant effects on housing affordability and rental availability in Cape Town. While such platforms provide income opportunities for hosts and increase tourism flexibility, concerns persist regarding professional hosting, housing stock diversion, and price inflation in residential areas. In this assignment, you will model the probability that an Airbnb listing in Cape Town is not fully booked over the next 30 days.

You are provided with `listings.csv`, containing (curated and filtered) Airbnb listing data for Cape Town from 2025. This file must be read into `R` as is, i.e. the spreadsheet may not be edited at all. The feature descriptions are given in `variable_descriptions.csv`. The target variable is `fully_booked_30`, which indicates whether or not a listing had zero availability over the next 30 days at time of data collection. **The positive/target class to use throughout this exercise is 'no'**, i.e. we are interested in identifying properties that are currently expected to be vacant for at least one night over the next month.

The data have already been cleaned and are ready for modelling. Although an extensive exploratory data analysis is not required, a brief section on this may be included. However, if you do so, your discussion should focus only on aspects that pertain to this exercise. You do not *have* to use all the features for the modelling, but if you omit any, this must be clearly communicated. If you want to (not necessary), you may perform additional preprocessing and/or feature engineering, as long as you **clearly document and report** what you did.

You are also provided with an unlabeled test set (`testing.csv`) for which predictions are to be uploaded (see Section 5).

All models are to be evaluated and compared according to 10-fold cross-validation (CV). The final evaluation metric that will be used is log-loss, aka binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right],$$

where $y_i \in \{0,1\}$ is the true label and $p_i = \hat{\Pr}(y_i = 1|\mathbf{x}_i)$ for observation $i = 1, \ldots, N$. To be clear: $y = 1$ if `fully_booked_30` is 'no'. Note that lower $\mathcal{L}$ is better.

Section 1 – Introduction and Data Preparation [10 marks]

Provide a brief introduction to the report – a reader should be able to comprehend the task at hand without having seen this set of instructions.

Include any data exploration/descriptions and decisions regarding the cleaning or manipulation of the data here.

Section 2 – Model Training [30 marks]

Using the full training set, fit the following models, tuned according to 10-fold CV. Be sure to clearly state/show your hyperparameter search space and report your final hyperparameter selection.

1. First fit a 'vanilla' logistic regression. Which variables appear notably significant/non-significant in this model?

   Now apply $L1$-regularisation. Which variables were removed/retained?

2. A random forest.

3. Any boosted tree model (gbm, xgBoost, LightGBM,[1] etc.).

Section 3 – Model Evaluation and Selection [30 marks]

a) Using your CV predictions, report the following metrics for each model in a single table: Log-loss, AUPRC, Recall, Specificity, Precision, F1 Score & Accuracy (in that order).

   Also provide a visual illustration of these results.

   For the metrics that require a predicted label, apply the decision rule

$$\hat{y}_i = \begin{cases} 1 & \text{if } \hat{p}_i \geq \tau \\ 0 & \text{if } \hat{p}_i < \tau \end{cases},$$

   where τ is the proportion of training observations belonging to the target class. (15)

b) Briefly compare the models and select (with motivation) your final model. (5)

c) Plot the CV ROC curve for your selected model. Indicate on the curve the point corresponding to τ, and add this value as text next to the point. (10)

Section 4 – Interpretation [20 marks]

For the selected model:

a) Provide variable importance plots and briefly discuss. (10)

b) Provide at most four partial dependence plots (your choice of how many) and interpret them in context. (10)

   If the logistic regression was your final model, replace a) and b) with a comprehensive interpretation of the retained variables' coefficients.

[1] The documentation can be found here.

Section 5 – Prediction [10 marks]

Use your selected model to predict the `fully_booked_30` probabilities for the observations in `testing.csv`. Write your predictions to a .csv file as follows:

- The file must only contain ONE column, namely the predicted probabilities, i.e. just a column of numbers between 0 and 1.
- Remember, these must represent $\hat{\Pr}(\texttt{fully_booked_30} = \text{'no'})$.
- Do NOT include a column header.
- There should be NO blank cells to the left or above this column.
- The order of the predictions must correspond to the order of the observations in `testing.csv`.
- The file name must be your student ID, all uppercase. E.g. `STDNUM001.csv`.

You will only lose marks if you fail to adhere to these instructions (check your file before submitting!), or if your predictions are nonsensical or seemingly random. The ranked log-loss results will be posted, although this is purely for fun/bragging rights and will have no bearing on your assessment mark.

$\mathcal{END}$

TOTAL MARKS = 100